PHY-112 | Principles of Physics-2

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Spring 2025 | Class #7

Department of Mathematics & Natural Sciences

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Recap of the previous class!

What we studied in Class #5

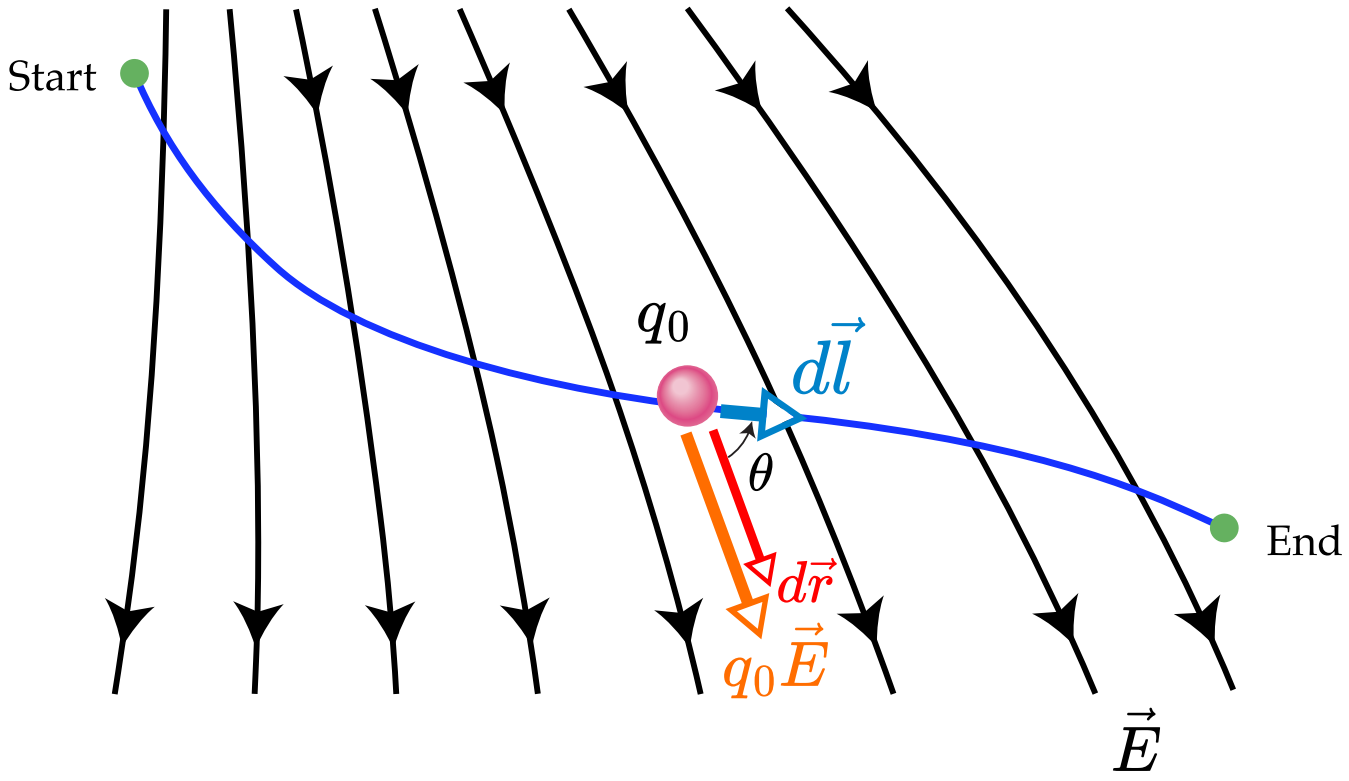
Refer to Class #5 slides for details!

- Application of Gauss's Law: Cylindrical Symmetry, i.e., Line Charge
- Application of Gauss's Law: Planar Symmetry, i.e., Plane Charge
- Application of Gauss's Law: Spherical Symmetry, i.e., Volume Charge (Would be provided in a recorded video | Class 6)

To Work or not to Work!

Work Done due to Electrostatic Force

When charges move in presence of fields



Work Done due to Electrostatic Force

When charges move in presence of fields

When a charge moves or is moved in the electric field, work is done by or against the Coulomb force to accelerate it, thus displacing it. This work is stored as potential energy in the system.

$$W = \int dW = \int \vec{F}_E \cdot d\vec{l}$$

$$= q_0 \int_{\text{initial}}^{\text{final}} \vec{E} \cdot d\vec{l} = q_0 \int_{\text{initial}}^{\text{final}} E dr;$$

where $dr = dl \cos\theta$.

Types of Work Done due to Electrostatic Force

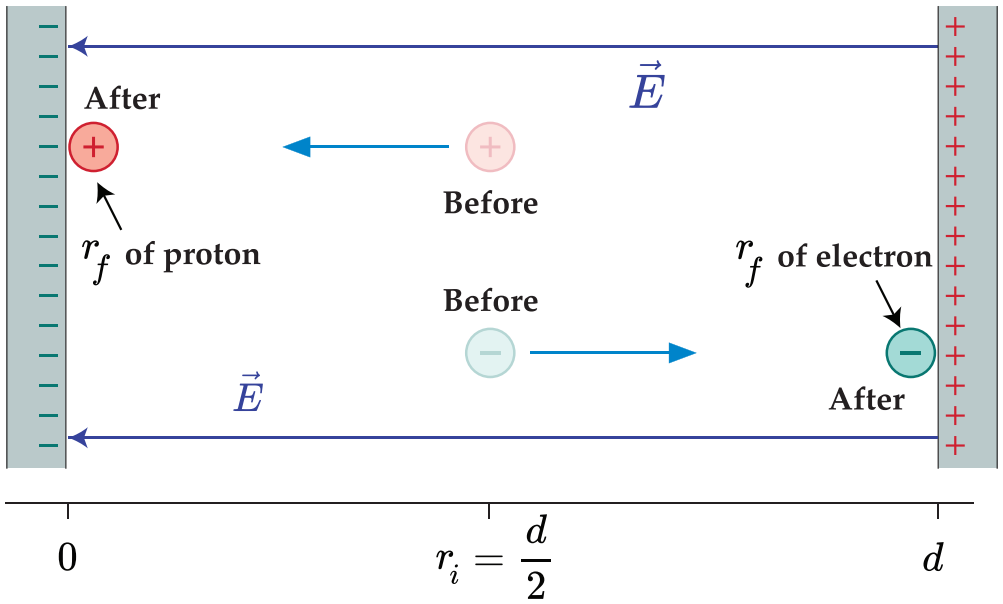
Psst! The system is the charge

- Charges **moves** on its own due to Coulomb Force $\vec{F}_E \longrightarrow$ Positive Work (By the System)
- Charges are **moved** using external force $\vec{F}_{\text{ext}} \longrightarrow$ Negative Work (On the System)

Incepting Ideas (1)

Hint: Follow where the dot product leads You

Q: A parallel-plate capacitor with a 2.0 mm spacing produces a uniform $\vec{E}$-field of intensity 2.85×10^5 N C^{-1}. First, a proton and then an electron are released from rest midway between the capacitor plates. Assume the motion takes place in a vacuum.



- What is the work done on each particle during this process? Is it done by the system or on the system?

Energy with Electric Fields

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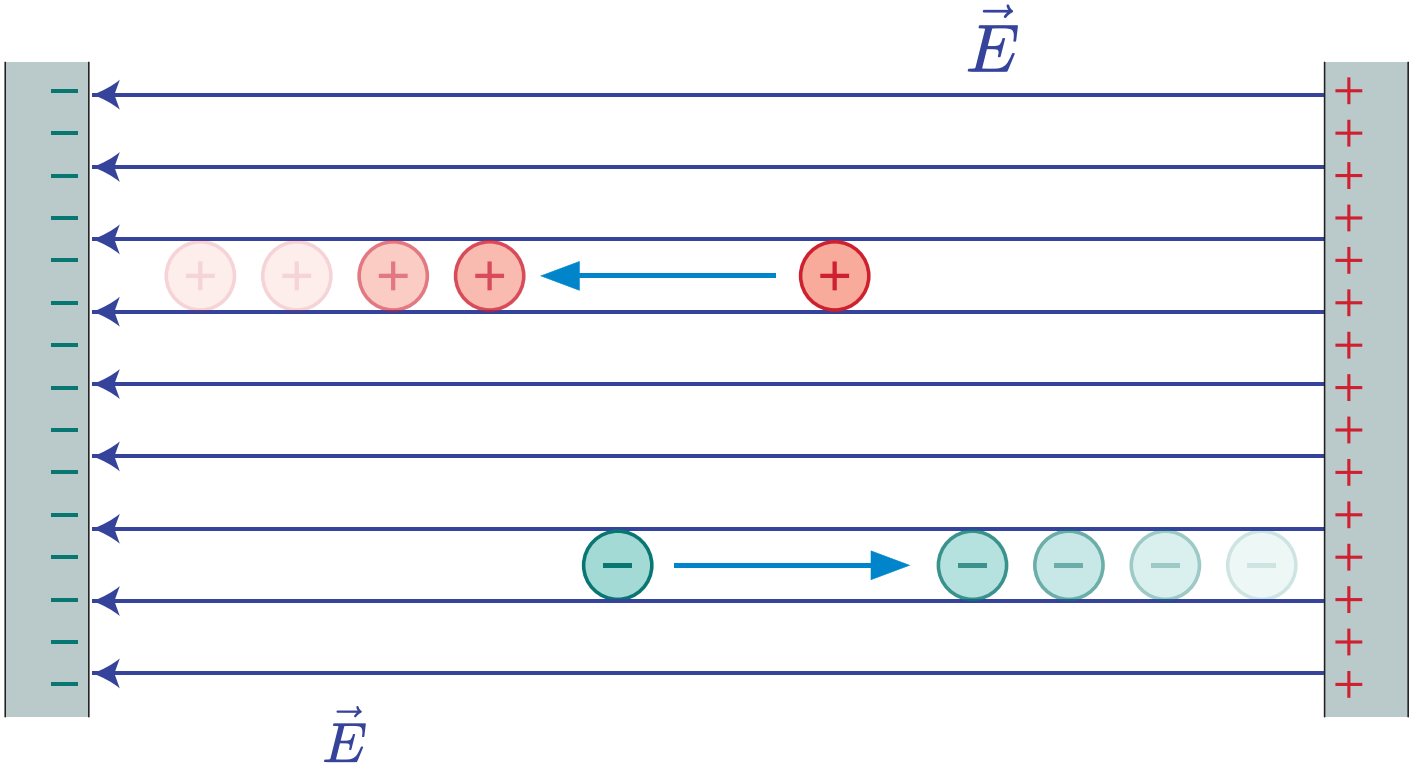
Electric Potential Energy

Energy that comes out of interaction with fields



Electric Potential Energy

Energy that comes out of interaction with fields

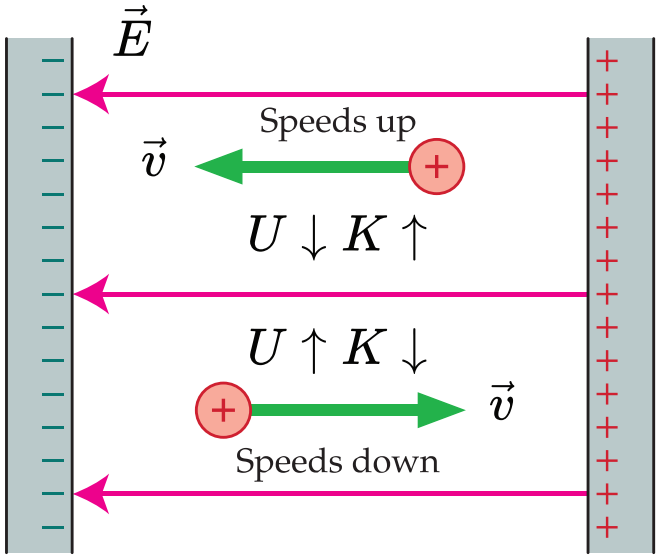
- Sheet Charges are good examples of sources that produce Non-Uniform $\vec{E}$-fields
- The Potential Energy of a Charge in a Uniform Electric Field

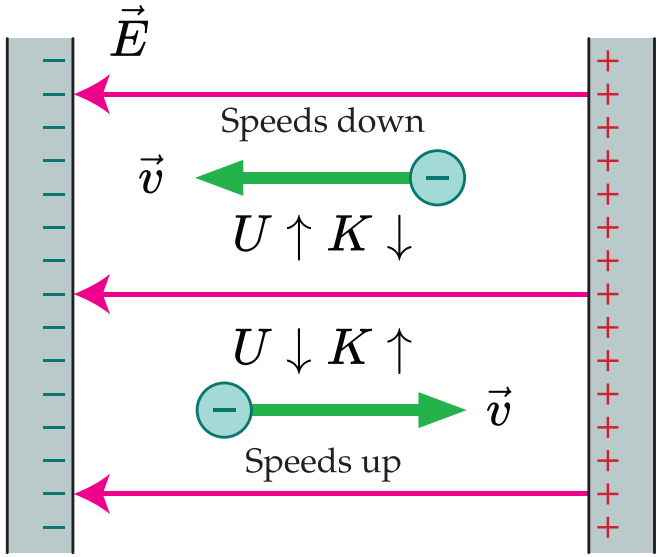
$$\Delta U_{\text{elec}} = -W_{i\to f} = -q_0 \int_{\text{initial}}^{\text{final}} \vec{E} \cdot d\vec{l} = -q_0 \int_{\text{initial}}^{\text{final}} E dr;$$

where $dr = dl \cos\theta$.

Energy Exchange of a Charged Particle in $\vec{E}$-field

Pingpong match between U and K





Energy Exchange of a Charged Particle in $\vec{E}$-field

Pingpong match between U and K

During Work Done **by** the system:

$$q_0 > 0 \longrightarrow \begin{cases} \Delta U < 0; \text{ Potential Energy is lost} \\ \Delta K > 0; \text{ Kinetic Energy is gained} \end{cases}$$

$$q_0 < 0 \longrightarrow \begin{cases} \Delta U < 0; \text{ Potential Energy is lost} \\ \Delta K > 0; \text{ Kinetic Energy is gained} \end{cases}$$

Energy Exchange of a Charged Particle in $\vec{E}$-field

Pingpong match between U and K

During Work Done **on** the system:

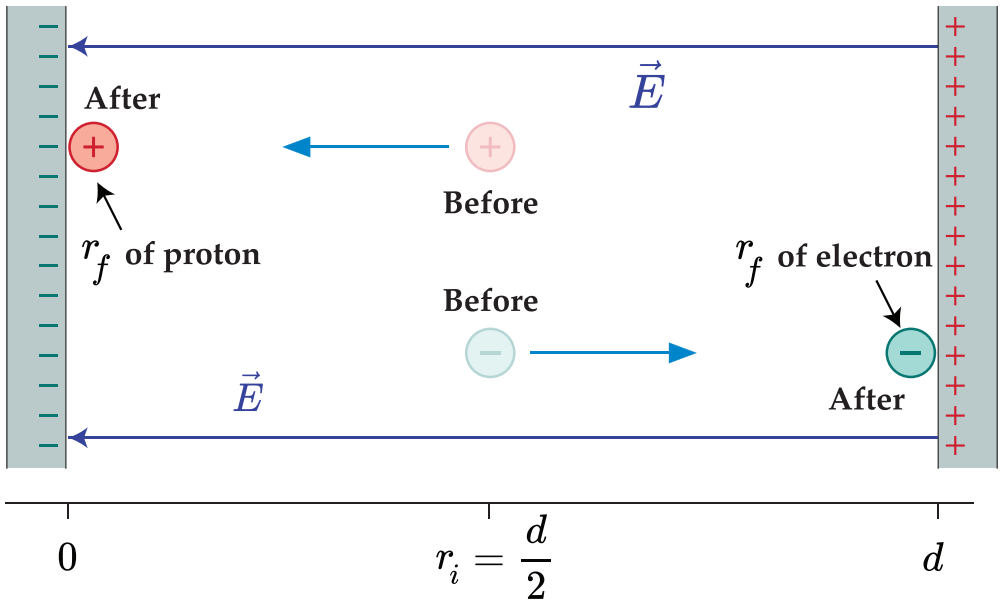
$$q_0 > 0 \longrightarrow \begin{cases} \Delta U > 0; \text{ Potential Energy is gained} \\ \Delta K < 0; \text{ Kinetic Energy is lost} \end{cases}$$

$$q_0 < 0 \longrightarrow \begin{cases} \Delta U > 0; \text{ Potential Energy is gained} \\ \Delta K < 0; \text{ Kinetic Energy is lost} \end{cases}$$

Incepting Ideas (2)

Hint: Apply the Energy conservation principle

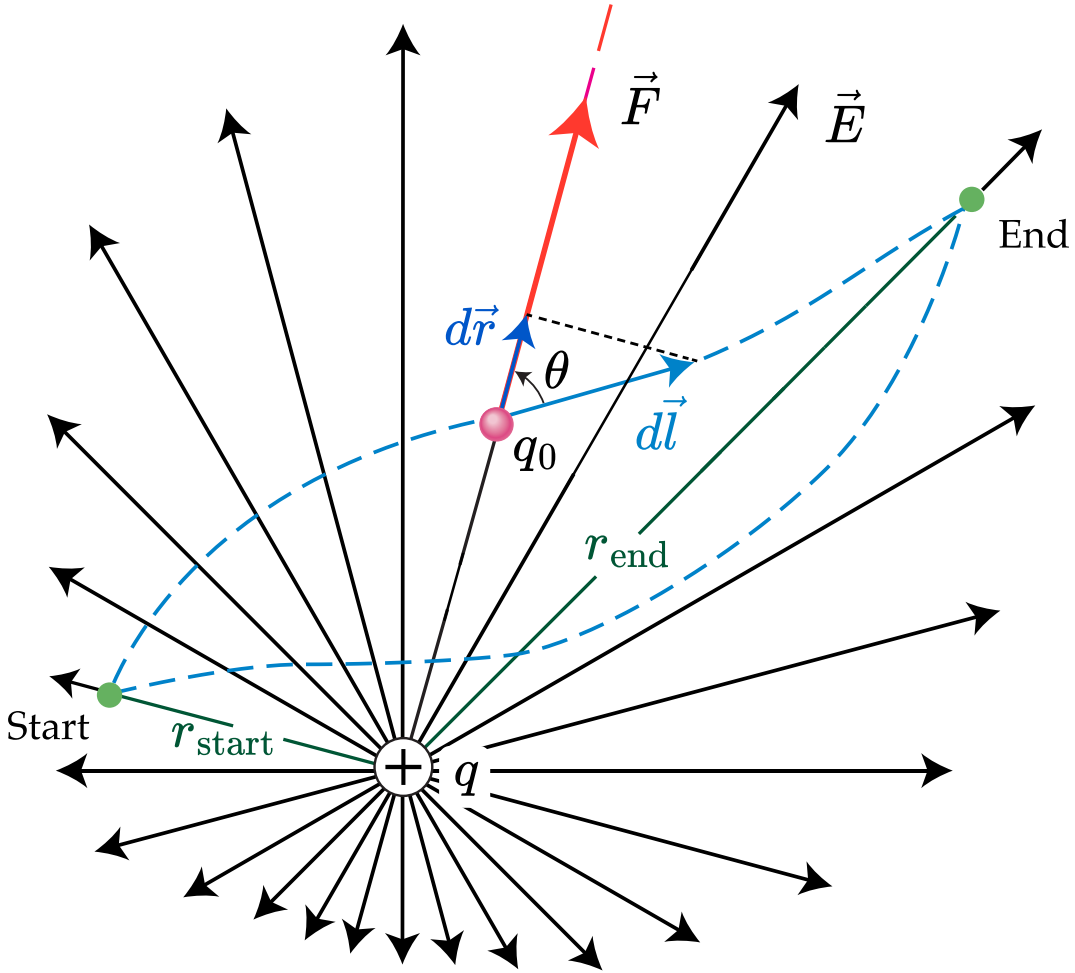
Q: A parallel-plate capacitor with a 2.0 mm spacing produces a uniform $\vec{E}$-field of intensity 2.85×10^5 N C^{-1}. First, a proton and then an electron are released from rest midway between the capacitor plates. Assume the motion takes place in a vacuum.



- Do the particles gain or lose potential energy? What about their kinetic energies? Calculate the amounts.

Electric Potential Energy

With point source



Electric Potential Energy
With point source

- Point Charges are good examples of sources that produce Non-Uniform $\vec{E}$-fields
- The Potential Energy of a Charge in a Non-Uniform Electric Field

$$\Delta U_{\text{elec}} = -q_0 \int_{\text{initial}}^{\text{final}} \vec{E} \cdot d\vec{l} = -q_0 \int_{\text{initial}}^{\text{final}} E dr$$

$$= -\frac{q_0 q}{4\pi\epsilon_0}\left(\frac{1}{r_{\text{initial}}} - \frac{1}{r_{\text{final}}}\right)$$

$$= U_{\text{final}} - U_{\text{initial}}$$

The Potential Energy of Point Charges

You have seen this before

Physically, electric potential energy represents the work required to transport a charge **from infinity**, where the electric field is considered to be zero, to a specific point r **within an electric field**.

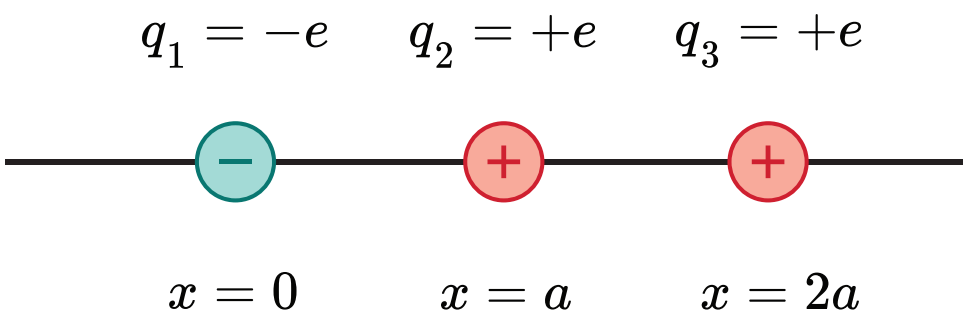
$$U_{\text{elec}} = -W_{\infty} = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r}$$

Alternatively, electric potential energy can also be defined as the work required to **move a charge from a finite distance** r (from the source) **to infinity**, where the electric field is considered to be zero, with the sign of the work being reversed.

Incepting Ideas (3)

Scalars can be added like numbers

Q: Two point charges are at fixed positions on the x-axis, $q_1 = -e$ at $x = 0$ and $q_2 = +e$ at $x = a$. Take $a = 2$ cm.

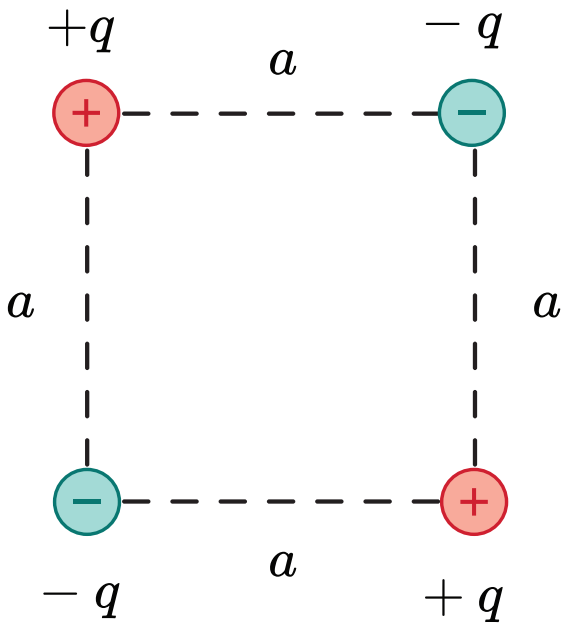


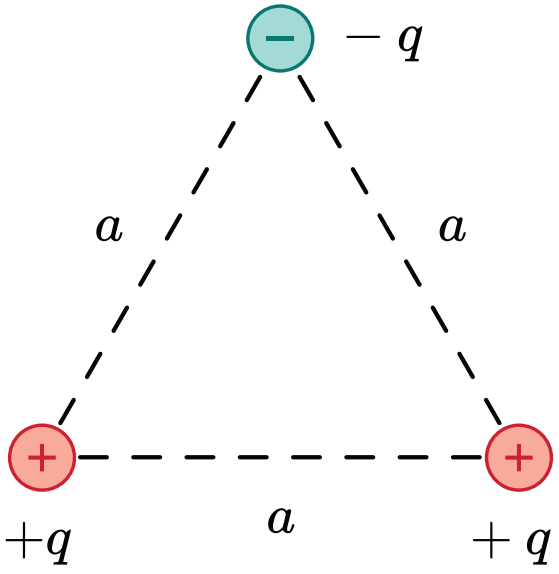
- Find the work that must be done to bring a third point charge $q_3 = +e$ from infinity to $x = 2a$.
- Find the total potential energy of the system of three charges.

Incepting Ideas (4)

Scalars can be added like numbers

Q: How much work is required to set up the arrangement in this square and triangle formation? The particles are initially infinitely far apart and at rest. Take $a = 2$ cm and $|q| = 2.30$ pC.





Incepting Ideas (5)

Practice! Practice! Practice!

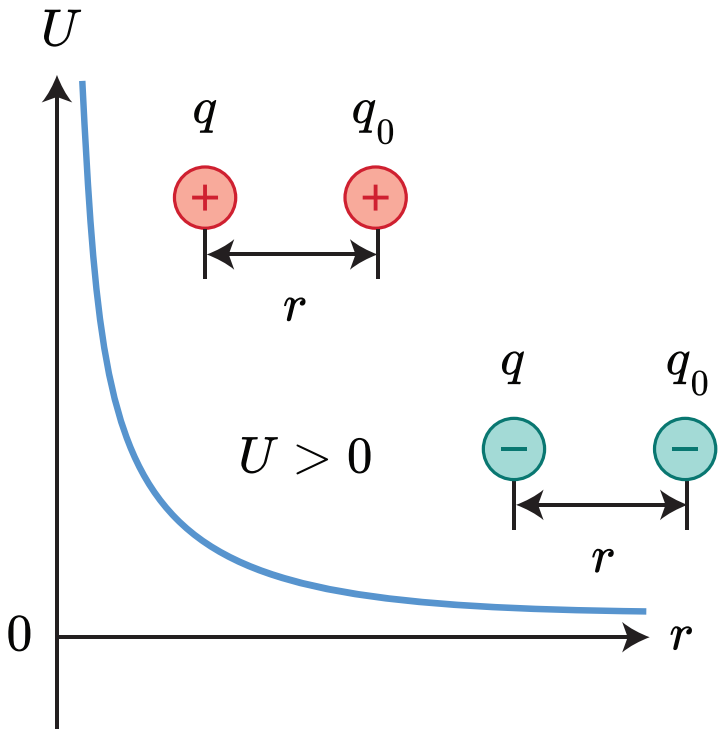
Some Practice Problem to try at home on today's topics

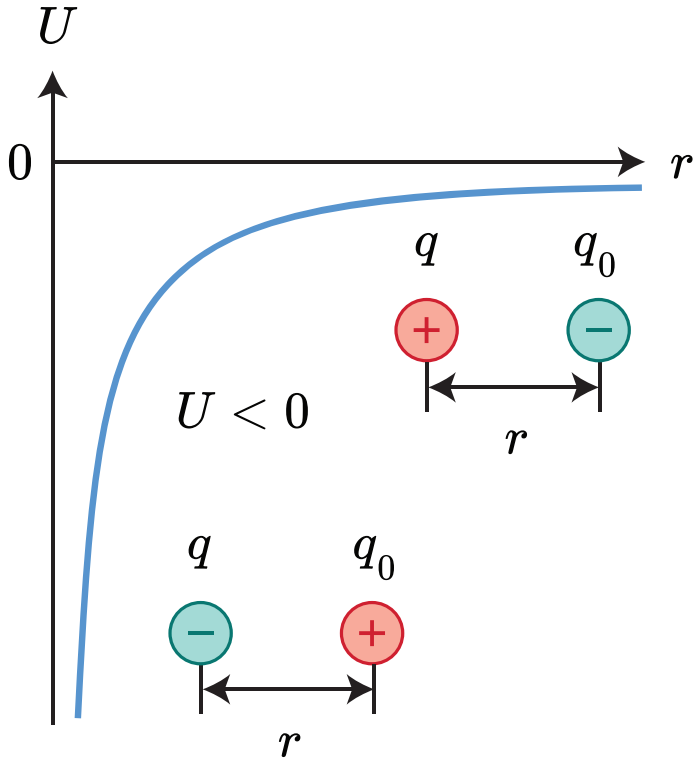
Exercise Problem 23.1, 23.2, 23.8, 23.10, p-801 | Young-Freedman
Sample Problem 24.01, p-689, 24.06, p-704 | Resnick-Halliday

Physics? Or Magic?

The Potential Energy of Point Charges

Physics is superior than magic!





Physics, when not properly explained, can be indistinguishable from magic.

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That is it for today!